

NAME: \_\_\_\_\_

PERIOD: \_\_\_\_\_

DATE: \_\_\_\_\_

## One Difference, Not Two Separate Intervals

AP Statistics · Topic 6.8 (CED 3.10) · B: 2026-12-11 · E: 2027-01-22

### [STAR] TODAY'S PROBLEM

Town Oak has 2,200 households and Pine 1,800. Independent SRSs find 77 of 140 Oak households and 48 of 120 Pine households use curbside composting. Let  $p_O - p_P$  mean Oak minus Pine.

**Elena:** “The two-sample  $z$ -interval is  $(0.030, 0.270)$ .”

**Devon:** “Subtract the farthest endpoints of separate 95% intervals. That gives  $(-0.020, 0.320)$ , so zero is plausible.”

#### Household samples

| Town | Use | Other | n   | N    |
|------|-----|-------|-----|------|
| Oak  | 77  | 63    | 140 | 2200 |
| Pine | 48  | 72    | 120 | 1800 |

#### FIRST TAKE — before the video (5 min)

Which method gives the requested 95% interval for  $p_O - p_P$ ? Cite one design or numerical detail.

#### [BOOK] BUILD ONE INTERVAL FOR THE DIFFERENCE (keep on the page)

For two independent random samples, estimate  $p_1 - p_2$  with a two-sample  $z$ -interval:  $(\hat{p}_1 - \hat{p}_2) \pm z^* \sqrt{\hat{p}_1(1 - \hat{p}_1)/n_1 + \hat{p}_2(1 - \hat{p}_2)/n_2}$ . Verify independent random samples or random assignment,  $n_i \leq 0.10N_i$  when sampling without replacement, and at least 10 observed successes and failures in each group. For an interval, use each group's sample proportion separately; because the samples are independent, estimated variances add before taking the square root. Two separately constructed  $C\%$  intervals do not automatically combine into one  $C\%$  interval for  $p_1 - p_2$ .

**Finish after the video:**

**DOK 1** (a) Define  $p_O$  and  $p_P$  and name the interval procedure for  $p_O - p_P$ .

**DOK 2** (b) Check independent samples, both 10 percent conditions, and all four counts. Compute the estimate, SE, margin of error, and 95% interval.

**DOK 3** (c) Choose the warranted method. Use design, counts, variance structure, and zero's location. Explain why separate 95% intervals do not produce the requested coverage and how Devon's result misleads.

**Frame:** The warranted method is \_\_\_\_\_ because the data come from \_\_\_\_\_, and the four counts are \_\_\_\_\_.

**Frame:** Devon's interval misleads a reader because \_\_\_\_\_; as a result, zero appears \_\_\_\_\_ even though \_\_\_\_\_.

**EXIT — turn this in**

One thing the video changed about my first take: \_\_\_\_\_